

REVIEW

Therapeutic Strategies Combining Antiplatelet With Anticoagulant Actions: Heparin Proteoglycan Mimetic APAC as a Model

Axelle Y. Kern, Annukka Jouppila, Hanna Pitkänen^{ID}, Riitta Lassila

ABSTRACT: Current antithrombotic therapies (combining aspirin, purinergic P2Y₁₂ inhibitors, and anticoagulants) target both platelet and coagulation pathways to reduce ischemic events in high-risk cardiovascular patients. Despite efficacy, these regimens are limited by increased bleeding risk, as evidenced by multiple clinical trials. Suboptimal pharmacodynamics and nonindividualized treatment approaches complicate the therapeutic success. Pediatric cardiac conditions and severe thrombophilia underlying arterial and immune thrombosis pose special challenges. The dual actions aim at addressing these gaps by especially attenuating arterial platelet-rich thrombi and safety concerns. We highlight the need for improved antithrombotic strategies across diverse patient populations. To potentially overcome the limitations, we present antiplatelet and anticoagulant (APAC), a heparin proteoglycan mimetic integrating APAC effects. APAC, administered locally or intravenously, targets VWF (von Willebrand Factor) and thrombin, effectively inhibiting platelet-fibrin deposition, under high shear rate flow conditions over surfaces of collagen and TF (tissue factor). Preclinical models show that APAC reduces atherosclerotic and stenotic lesions, inflammation, and ischemia-reperfusion injury without bleeding propensity. These findings position APAC as a promising candidate for safe, translational dual antithrombotic. Early clinical trials demonstrated the safety of APAC under both systemic and local administration. A phase 2 clinical study in peripheral arterial disease is underway, and another is planned for arteriovenous fistula patency for end-stage renal patients.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: antithrombotic agent ■ blood coagulation ■ hemorrhage ■ platelets ■ thrombosis

Basic science of thrombosis and hemostasis has demonstrated the key roles of platelets in interaction with an injured vessel wall, especially in arterial circulation. Exposed VWF (von Willebrand Factor) and collagen initiate arterial thrombosis with blood flow impacting the outcome critically. The higher the shear forces are, the more platelets and VWF will contribute, while coagulation and fibrin formation participate more at the lower venous shear forces¹ (Figure 1). In areas of peak stenosis and microvasculature, the shear rate increases over several thousand s⁻¹, reducing the formation of thrombin and fibrin and resulting in platelet-rich thrombi. The blood content of erythrocytes and VWF also positively impacts hemorheology by enhancing shear forces upon vessel walls.^{1,4}

In venous clot models, thrombin generation and activity together with platelet releasate are influenced

by the geometry of the thrombus, with areas of both loosely and densely packed platelets, and chemical stagnation.⁵ In arterial circulation, under high shear rate conditions (over 600 s⁻¹), GPIIb/IIIa (glycoprotein) mediates the primary adhesion of platelets to VWF and VWF-bound collagen, whereas platelet adhesion relies more on integrin $\alpha_{IIb}\beta_3$ -fibrinogen binding under lower shear rates (below 600 s⁻¹).⁶ These wall shear rates and immobilized ligands vary in complex stenotic lesions, and in the flow divider areas, and develop during thrombus growth. Moreover, various plasma-related thrombin activities impact the localized stabilization of the clot, mostly related to the distal part of the thrombus.⁷

Blood flow over stenotic and branching arteries downstream of the lesion trigger local hemorheological conditions, which impact the composition of thrombus

Correspondence to: Riitta Lassila, MD, PhD, Faculty of Medicine, Research Program in Systems Oncology, University of Helsinki, Helsinki, Finland. Email riitta.lassila@kolumbus.fi

© 2026 American Heart Association, Inc.

Arterioscler Thromb Vasc Biol is available at www.ahajournals.org/journal/atvb

Nonstandard Abbreviations and Acronyms

APAC	antiplatelet and anticoagulant
aPTT	activated partial thromboplastin time
ASA	acetylsalicylic acid
AVF	arteriovenous fistula
CCL2	C-C motif chemokine ligand 2
CFT	clot formation time
CIPA	collagen-induced platelet aggregation
CT	coagulation time
HEP-PG	heparin proteoglycan
IL	interleukin
IRI	ischemia-reperfusion injury
PAOD	peripheral arterial occlusive disease
PF4	platelet factor 4 (CXCL4)
TF	tissue factor
TFPI	tissue factor pathway inhibitor
UFH	unfractionated heparin
VWF	von Willebrand Factor

Highlights

- Dual-antiplatelet and anticoagulant strategies aim at improving antithrombotic treatments while maintaining safety for patients at risk.
- Antiplatelet and anticoagulant (APAC), a heparin proteoglycan mimetic, carries both antiplatelet and anticoagulant properties by targeting von Willebrand Factor and thrombin.
- APAC in experimental models reduces atherosclerotic plaque formation, thrombosis, inflammation, and ischemia-reperfusion injury without bleeding.
- Safety studies support APAC's clinical programs in thrombogenic patients without increased bleeding risks.

formation.¹ At the low flow and recirculation zones, coagulation activity prevails, and fibrin is formed, providing the target for thrombin inhibition. In all, mixed thrombi are formed.^{1,5} Moreover, the spatial regulation of loosely

bound platelets versus activated and fibrin-entangled platelets has been depicted in thrombosis. Red blood cells, by interacting with both platelets and fibrin, consolidate the clot by causing polyhedrocytes, alike cobble stones, securing hemostasis.^{7,8} Hemolysis, thromboxane, and thrombin generation and sickling abnormalities of red blood cells play additional mechanisms to accelerate thrombosis and provide resistance to fibrinolysis.^{9–11}

In the case of inflammation or poor regulation of vasoactivity or vasoconstriction, platelets, leukocytes, and endothelial cells release microcellular or extracellular

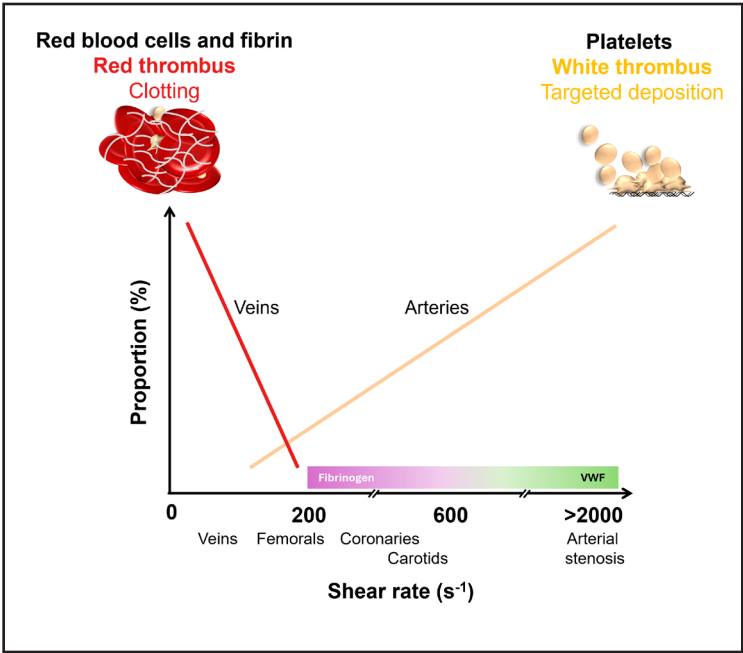


Figure 1. Influence of shear rate on formation and structural composition of thrombus. Shear forces play an important role in both the kinetics and architecture of thrombus formation.² Low shear rate conditions (under 200 s⁻¹) in veins favor enhanced activation of the coagulation over platelets, resulting in fibrin-dense and red blood cell-dense (red) clots. In contrast, under high shear rate blood flow conditions (exceeding 600 s⁻¹) in diseased arteries, platelet adhesion and activation are predominantly mediated by VWF (von Willebrand Factor) and collagen. This condition drives the formation of platelet-rich (white) thrombi. Platelet aggregation in arteries with lower shear rates (200–600 s⁻¹) is more fibrinogen-dependent. While the human average shear rates are 135 s⁻¹ for the femoral,³ 260 to 500 s⁻¹ for the carotid, and 100 to 350 s⁻¹ for the coronary arteries,² stenosis or vasoconstriction enhances the local shear rates over 2000 s⁻¹.

vesicles. These mediate thromboinflammatory conditions in situ and downstream. Microvesicles have an ability to bind to fibrin surfaces¹² and promote platelet procoagulant properties and cargo messaging to distant vascular sites.¹³ Atherosclerotic lesions, either with endothelial activation, erosion, or rupture, expose TF (tissue factor), enhancing thrombin generation and fibrin formation in situ.¹⁴ Microvesicles bound to fibrin possess procoagulant activity by exposing negatively charged phospholipids, that is, phosphatidylserine, on their external membrane, offering a surface for tenase and prothrombinase complexes, alike activated platelets as well.^{12,15} The phospholipid exposure and presence of TF support thrombin generation downstream of platelet aggregation.¹⁶

First and last, the endothelium, especially with its glycosaminoglycans, proteoglycans, and other antiplatelet and anticoagulant contributors, including the thrombomodulin-protein C axis and the TFPI (TF pathway inhibitor), controls hemostasis.^{17–20} Thromboinflammation may shed glycosaminoglycans from the vessel surfaces, via degradation or derangement by heparanases.²¹ This is relevant in cancer and other thromboinflammatory conditions. There, the revealed thrombogenic vessel surfaces due to proteolysis or disruption continue the rolling of the vicious loop.

PATHOGENETIC IMPLICATIONS OF THROMBOSIS FOR ANTITHROMBOTIC THERAPY

When antithrombotic agents are needed, either to prevent arterial thrombosis from recurring (secondary prevention) or to manage acute thrombosis, the treatment between arteries and veins differs. For prevention, acetylsalicylic acid (ASA), inhibiting thromboxane A₂ formation, or clopidogrel, inhibiting ADP-induced platelet aggregation, is mostly used. Ticagrelor or prasugrel, the other 2 P2Y₁₂ receptor inhibitors, has targeted indications after unstable angina or myocardial infarction, that is, acute coronary syndromes,²² but they are not in the arsenal for other arterial locations. Particularly, stroke, whether due to an embolus from the left atrium or aortic arch or in situ platelet-driven thrombi, is susceptible to hemorrhagic complications in association with antithrombotic therapies. Some compliant patients may be poor responders to ASA^{23,24} or clopidogrel,²⁵ heightening the risk of recurrent thrombosis. This calls upon a switch to either prasugrel or ticagrelor,^{26,27} with more potent prolonged systemic effects, complicating hemostasis if interventions are needed. The diminution of ASA or clopidogrel effect is also associated with obesity/insulin resistance²⁸ or with COX-1 (cyclooxygenase-1) and platelet glycoprotein VI polymorphism.²⁹

Nowadays, gradually shorter treatment periods of dual-antiplatelet therapy are advocated for coronary

lesions after stenting because, unsurprisingly, the combination of antithrombotic drugs challenges hemostasis, also in other arterial manifestations.³⁰ However, even patients with hemophilia having high cardiovascular risks and FVIII or FIX levels of only 5 IU/d may transiently use these therapies, depending on the indication.³¹ Because the tolerance for dual-antiplatelet therapy is highly individual, it calls for generalized risk assessment for bleeding complications. As one example, in association with acute coronary syndrome needing acute coronary artery bypass grafting, ASA alone was safer (beyond 2 months) but as efficient as ASA and ticagrelor combined during 1-year follow-up.³²

For treatment of acute arterial thrombosis, combined antiplatelet(s) and anticoagulant benefit natural fibrinolysis to resolve the arterial clot or support invasive balloon angioplasty with or without stenting. Mechanistically, this combination may well tackle the antifibrinolytic role of both platelets and thrombin. Platelets contain FXIII, plasminogen activator inhibitor-1, and antiplasmin and promote thrombin generation, which, in turn, stimulates TAFI (thrombin-activatable fibrinolysis inhibitor).^{7,33}

The combined therapy is needed for the short term, a week to 3 months. For prevention or long-term treatment, depending on the nature of thrombosis, only a single strategy is deemed safe in several indications. One exclusion is peripheral arterial occlusive disease (PAOD), which often carries the highest morbidity and mortality of the cardiovascular diseases, being also highly thrombogenic.³⁴ Rivaroxaban, an FXa inhibitor at a low dose together with ASA, provided benefits both after high-risk acute coronary syndrome (COMPASS trial [Cardiovascular Outcomes for People Using Anticoagulation Strategies]) and PAOD (VOYAGER PAD study [Vascular Outcomes Study of ASA Along With Rivaroxaban in Endovascular or Surgical Limb Revascularization for Peripheral Artery Disease]), yet, with the price of increased bleeding tendency.^{35,36}

Some additional severe thromboses may need intensified antithrombotic treatments with watchful stewardship for hemostasis. These specific situations include pediatric thrombosis, cardiac poor outflow syndromes, or heart surgery, and their management with a parenteral thrombin inhibitor. Moreover, the combined thrombophilia and cardiovascular comorbidities with high demands for efficacy will benefit from combined antithrombotic strategies.³⁷

Thrombin Inhibitors in Children

Until recently, unfractionated heparin (UFH) has been used as the only anticoagulant for pediatric extracorporeal membrane oxygenation and ventricular assist device populations.^{38,39} Unfortunately, UFH has important limitations, especially in association with the management of neonates and young infants. UFH needs optimal

antithrombin activity for the effects but demonstrates variable pharmacokinetics in children. UFH also binds to other plasma proteins and endothelial cells. The concentrations of the plasma proteins depend on age and underlying endothelial and clinical conditions, resulting in unpredictable UFH responses.^{40–42} These limitations have encouraged the use of alternative intravenous direct thrombin inhibitors, namely, bivalirudin, argatroban, and dabigatran, which can inhibit thrombin both in circulation and when fibrin-bound, decreasing both platelet activation, and with the feedback loop the stability of coagulation to promote thrombolysis.^{37,43}

Clinical Needs in Pediatric Anticoagulation

The pediatric hemostatic system differs markedly from that of adults, making the cause of thrombosis and the use of anticoagulants unique in this population. Children with univentricular hearts are particularly prone to adverse thromboembolic events.⁴⁴ Fontan circulation is the final palliation for patients with 1 functional ventricle or 2 incompetent ventricular pumps and severe congenital heart disease. During the first stage of Fontan palliation for neonates, systemic and pulmonary blood flow circulations are balanced using surgical shunts, either Blalock-Thomas-Taussig or right ventricle-to-pulmonary artery shunt.⁴⁵ Postoperatively, the patients are anticoagulated with low-molecular-weight heparin before being switched to ASA. Despite this, the in-hospital mortality range for Blalock-Thomas-Taussig shunts varies from 5.2% to 13.6%.^{46–48}

Percutaneous patent duct stenting with adult coronary stents has been developed as a minimally invasive alternative to the Blalock-Thomas-Taussig shunt. ASA is started a day before the procedure, followed by UFH infusion for 24 hours postoperatively, continued with low-molecular-weight heparin, and finally maintaining ASA and clopidogrel.^{44,49}

Neonatal patients with shunts or stents need reassurance of patent pulmonary blood flow and would theoretically benefit from combined anticoagulant and antiplatelet therapy, especially during the acute postoperative phase. As shunt thrombosis increases morbidity and mortality, targeted antithrombotic options for these patients should be explored.

Thrombin Inhibitors in Adults

Adult patients with severe thrombophilia and acute arterial thrombosis, including patients with PAOD and mesenteric thrombosis, have benefited from combined antiplatelet and anticoagulation strategies.^{50–53} In patients with inoperable distal PAOD, lepirudin, a recombinant hirudin, likely prevented amputation by inhibiting thrombin-mediated platelet aggregation and propagation of thrombosis by binding to fibrin and offering a localized

antithrombotic action.^{43,51} Natural heparin proteoglycan (HEP-PG) mimetics could offer a novel antithrombotic strategy by reducing both VWF-dependent platelet and thrombin deposition, and modulating sterile inflammation, via complement regulation.

A NEW DUAL-ANTIPLATELET-ANTICOAGULANT MOLECULE

Heparin Proteoglycans as a Model for Antithrombotic

Mast cell-derived (rat serosal cavity) HEP-PGs inhibit collagen-induced platelet aggregation (CIPA) and reduce magnesium-dependent platelet deposition on collagen. Indeed, when introduced into blood perfused over a collagen surface or when immobilized on collagen, HEP-PG limited the platelet response to mere adhesion.^{54,55} These effects were particularly evident under high shear rate (1600 s^{-1}) VWF-dependent blood flow conditions. These in vitro observations were confirmed in vivo by local administration of HEP-PG to an anastomosed rat femoral artery, where HEP-PG protected against arterial thrombus formation without any other anticoagulant.⁵⁶ Conjugated UFH chains to albumin to create a semi-synthetic HEP-PG, inhibited platelet aggregation, and later, similar antithrombotic effects were demonstrated in various experimental models.⁵⁷ These findings have informed the development of a novel, locally acting dual antiplatelet and anticoagulant agent (APAC).

APAC as a New Strategy to Prevent Thrombotic Events

APAC comprises porcine UFH chains ($7\pm 2, \approx 17\text{ kDa}$ each) conjugated to a human serum albumin core (66 kDa), mimicking HEP-PG in structure and function as a local antithrombotic.^{54–56,58,59} Thereby, this highly negatively charged macromolecule possesses a strong affinity for positively charged coagulation and vascular proteins, enabling dual and localized antithrombotic actions. It also carries anti-inflammatory functions via complement regulation in vitro⁶⁰ and organ protection, preventing ischemia-reperfusion injury (IRI) of the kidneys and brain.^{61,62}

HEPARIN PROTEOGLYCAN MIMETIC APAC AS A POTENT ANTIPLATELET AGENT

Thrombosis Under Arterial Blood Flow Conditions

APAC prevents VWF- and thrombin-mediated platelet thrombosis in human blood. When perfused in a microfluidic flow device over both collagen and TF surface at arterial shear rate (1000 s^{-1}), APAC-anticoagulated

blood (10 $\mu\text{g/mL}$, without citrate or other anticoagulation) inhibited deposition of both platelets and fibrin,⁶³ unlike at low shear rate (200 s^{-1}) where fibrin formation was reduced. This clearly proved the antithrombotic action of APAC, supported also by the inability to generate serum, due to the absence of clotting.^{60,63}

Similarly, at high shear rate (1500 s^{-1}), under the anticoagulation with a strong FXa and thrombin inhibitor in the Total Thrombus-Formation Analysis System, APAC concentration-dependently (1–3 $\mu\text{g/mL}$) reduced occlusive thrombosis upon blood perfusion over collagen. Under citrate anticoagulation at a lower shear rate (600 s^{-1}) over a collagen and TF surface, APAC (3 $\mu\text{g/mL}$) also inhibited thrombosis but less than above. In contrast, UFH showed only a modest antithrombotic effect by prolonging occlusion time (34%–56%).⁵⁸ In all, both the strong external thrombin inhibition and the thrombogenic surface (collagen versus collagen and TF) impact the readouts and efficacy of APAC, illustrating the important role of thrombin.

Platelet Aggregation in Platelet-Rich Plasma

APAC is a potent and selective antiplatelet agent, primarily via the inhibition of CIPA. APAC concentration-dependently prolonged the lag time and decreased the slope of aggregation, reducing maximal and total aggregation in citrated platelet-rich plasma versus the vehicle.^{58,64} In contrast, UFH did not inhibit CIPA, even up to a 10-fold higher concentration.⁵⁸ Neither APAC nor UFH inhibited ADP-induced aggregation.^{57,58}

APAC's effects were also compared with cangrelor, a parenteral P2Y₁₂ receptor antagonist,⁵⁸ inhibiting both CIPA and ADP-induced aggregation. Coadministration of APAC potentiated the inhibition of CIPA (from 57% to 90%), while the ADP-induced aggregation remained unchanged.⁵⁸

Intracellular calcium mobilization assays using apixaban-treated platelet-rich plasma indicated that APAC specifically inhibited enzymatic thrombin and fibrillar collagen-triggered responses, unlike other agonists, that is, convulxin, ADP, U46619 (a TxA2 [thromboxane] analogue), or PAR-1- and PAR-4 (protease activated receptor)-specific agonists.⁶³

Overall, APAC shows strong antiplatelet activity by markedly reducing platelet and fibrin deposition on collagen and TF-coated surfaces under arterial shear rates, outperforming UFH.^{58,59} APAC selectively inhibits not only collagen (in platelet-rich plasma) but also thrombin-triggered (with washed platelets) platelet activation while allowing other signaling pathways. This APAC profile suggests an antiplatelet action targeting multifunctional interactions with the collagen-VWF-thrombin axis, rather than the earlier inhibitor of platelet glycoprotein VI, which lacks a known heparin-binding domain.⁶⁴ When the anticoagulant actions of spatially located heparin chains are

combined with this antiplatelet function, duality becomes a unique property for APAC.

APAC AS A POTENT ANTICOAGULANT

Whole Blood Coagulation

Dual antiplatelet and anticoagulant effects of APAC were also evidenced with intrinsic pathway-activated blood coagulation by rotational thromboelastometry InTEM⁵⁸ in citrated blood (Figure 2). APAC concentration-dependently prolonged (2–5-fold) both the coagulation time (CT), dependent on anticoagulation, and clot formation time (CFT), dependent on platelet procoagulant properties. The antiplatelet effects were also detected by a reduced α -angle and maximal clot firmness. At 3 $\mu\text{g/mL}$, APAC's anticoagulant effects already exceeded the maximal detection limits for CT and CFT completely in 23% of the samples, rendering maximal clot firmness unmeasurable. On the opposite, UFH (1–3 $\mu\text{g/mL}$) prolonged CT and CFT 30% to 70% less than APAC.⁵⁸

Overall, while APAC is a significantly more potent anticoagulant than UFH, it carries the unique inhibition profile on the platelet-related α angle and maximal clot firmness.

Plasma Coagulation

In spiked human plasma, APAC prolonged coagulation times more than an equivalent concentration of UFH. The anticoagulant and thrombin-mediated action of APAC clearly exceeded UFH when assessed with thrombin time rather than with activated partial thromboplastin time (aPTT).⁵⁷

Thrombin Generation

APAC concentration-dependently impaired thrombin generation in the calibrated automated thrombogram, by delaying both lag time and time to peak, while also reducing peak and endogenous thrombin potential in citrated plasma.⁵⁷ Though UFH had some anticoagulant activity in the presence of platelets, APAC (0.5 $\mu\text{g/mL}$) markedly inhibited thrombin generation.⁵⁷ APAC, already at 1 $\mu\text{g/mL}$, abolished thrombin generation, while some variables were still measurable at 1.5 $\mu\text{g/mL}$ of UFH.^{57,58} The intrinsic pathway-reflecting InTEM was downplayed, but this TF-driven calibrated automated thrombogram assay showed that APAC also inhibits the extrinsic pathway.

Various in vitro data indicate that APAC exerts a dual mechanism, as it integrates the inhibition of enzymatic thrombin with antiplatelet action via the collagen/VWF pathway while leaving other platelet-activating pathways ready for hemostasis. Compatibly, APAC does not interfere with ADP, TxA2, and thrombin receptor pathways, thus preserving platelet functionalities, including adhesion capacity. In comparison, APAC at low concentrations clearly surpasses UFH in the antithrombotic actions.

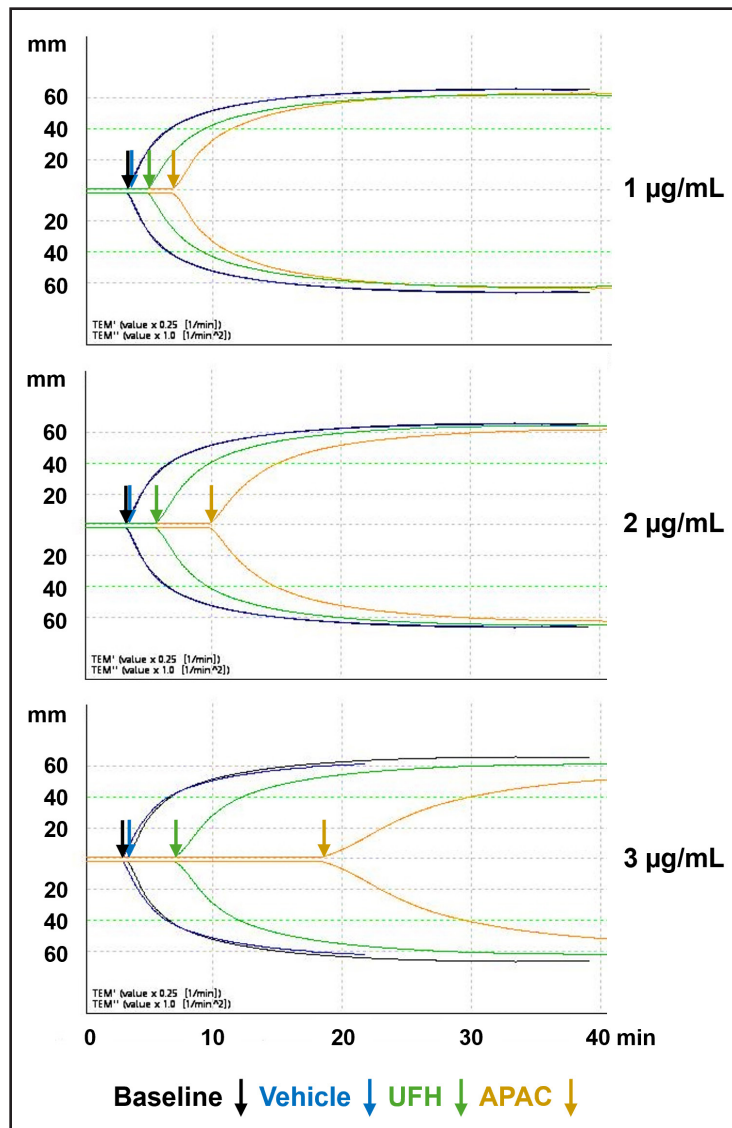


Figure 2. Antiplatelet and anticoagulant (APAC) prolongs blood coagulation more than unfractionated heparin (UFH).

Example of concentration-dependent (1–3 µg/mL) effects of APAC and UFH in citrated blood. The responses are measured by rotational thromboelastometry using intrinsic pathway-activated thromboelastometry (InTEM) in vitro. In addition to prolonged coagulation time, APAC importantly prolonged clot formation time and reduced clot firmness, indicating the platelet-inhibiting capacity. Unpublished tracing data originating from Jouppila et al.⁵⁸

NEUTRALIZATION OF APAC ACTIVITY BY PLATELET FACTOR 4

APAC's activities were both inhibited by PF4 (platelet factor 4) when assessed by CIPA in platelet-rich plasma, blood by InTEM, and plasma by aPTT and thrombin time. At 1 µg/mL, APAC inhibited CIPA (by 70%), while, after supplementation of PF4, the inhibitory effect markedly declined.⁵⁸ Similarly, APAC prolonged InTEM CT and CFT over 2-fold, but the addition of PF4 reduced this inhibitory activity by 37%.⁵⁸ Accordingly, APAC prolonged aPTT and thrombin time, while PF4 reversed these effects.⁵⁸ These observations suggest that APAC has a natural regulator for its antithrombotic actions in vivo.

According to coimmunoprecipitation and mass spectrometry analysis, APAC binds to PF4,⁶⁰ having anticoagulant reversal activity of APAC by a molar ratio of 8:1,⁵⁸ within the reported PF4/heparin ratio range.^{65,66} In vivo toxicology studies with APAC did not show any evidence of heparin-induced thrombocytopenia.⁶⁷

PRECLINICAL ANIMAL MODELS

The efficacy of APAC has been evaluated in several in vivo models described below, resulting in promising outcomes summarized in the Table. APAC exhibits potent antithrombotic action under arterial blood flow conditions (Figure 3). It selectively targets vascular injury sites

Table. Summary of Various Animal Experiments

Pathology	Species	Model	Injury site	Molecule (concentration)	Administration mode	Results	Reference
Acute arterial thrombosis	Baboons	Modified Folts model	Femoral artery and vein	APAC and UFH (2 mg)	Local	APAC inhibits occlusive thrombus formation for 2 h at the site of stenosis.	57
	Baboons	Collagen-coated graft connected to an extracorporeal AV shunt	Femoral artery and vein	APAC and UFH (0.5 mg)	Incubation of the graft prior to blood perfusion	APAC reduces platelet deposition, fibrin accumulation, and distal thrombus formation.	57
	WT mice	Photochemical endothelial injury	Carotid artery	APAC and UFH (0.3 mg/kg)	IV	APAC inhibits arterial thrombosis and platelet deposition, and prolongs the time to occlusion.	68
	WT mice	Collagen-induced thrombosis	Carotid artery	APAC and UFH (0.5 mg/kg)	IV	APAC's prophylactic administration reduces platelet accumulation.	68
Atherosclerosis	ApoE ^{-/-} mice	Collar-triggered stenosis with a high-fat diet	Carotid artery	APAC and UFH (0.2 mg/kg)	IV	Prophylactic APAC prevents early development of atherosclerosis and reduces inflammation.	69
Renal inflammation	Sprague-Dawley rats	Kidney IRI	Renal artery	APAC and UFH (0.13 mg/kg)	IV	APAC decreases the injury damages. Better effect in prevention than treatment.	61
Ischemic stroke	WT mice	tMCAO	MCA	APAC and UFH (0.5 mg/kg)	IV	Prophylactic APAC reduces cerebral IRI.	62

APAC indicates antiplatelet and anticoagulant; ApoE^{-/-}, apolipoprotein E knock out; AV, arterio-venous; IRI, ischemia-reperfusion injury; IV, intravenous; MCA, middle cerebral artery; tMCAO, the middle cerebral artery occlusion; UFH, unfractionated heparin; and WT, wild type.

(Figure 3A) while maintaining initial platelet adhesion to conserve primary hemostasis, as observed in mice where intravenous APAC inhibited platelet thrombus formation and TF activity at the injury site. As in in vitro human experiments, APAC also potently inhibited VWF-, collagen- and thrombin-mediated platelet aggregation, as well as fibrin formation (Figure 3B), effectively preventing thrombus growth in high-shear environments. Finally, APAC cools inflammation (Figure 3C), via indirect antithrombotic action, by reducing IRI and regulating the complement system at cell surfaces.

APAC Activity in Murine Blood In Vitro

Comparable with human platelets, APAC inhibited CIPA with murine platelets,^{58,61,68} unlike UFH, but, opposite to human, APAC inhibited also ADP-induced aggregation by 50% in vitro.⁶⁸

APAC and UFH inhibited coagulation with equal efficacy in mouse plasma.⁶² However, while APAC and UFH (0.5 μg/mL) prolonged thrombin time by 1.3-fold, APAC was 1.2- to 2-fold more potent at higher concentrations.^{62,68} This difference was less on aPTT.^{62,68} Overall, like humans, mouse plasma emphasizes the thrombin inhibition by APAC.

Distribution and Binding of Locally Administered APAC on Vascular Injury Site

APAC's seeking to surgically induced vascular injury sites was assessed in a porcine model of arteriovenous fistula (AVF) created between the femoral artery and

vein (Figure 4) and in balloon catheter denuded iliac and carotid arteries.⁷⁰ Streptavidin detected biotinylated APAC to cover the luminal area around the AVF, reached the internal elastic lamina in the femoral artery, and penetrated the deeper layers of the tunica media in the femoral vein.⁷⁰ Similar results were obtained in the balloon-denuded arteries.⁷⁰

In the 2 pig models of AVF and balloon angioplasty, APAC colocalized with vascular VWF and laminin (Figure 5).⁷⁰ Furthermore, the interaction between APAC and recombinant human VWF was confirmed in vitro by immunoprecipitation⁶³ studies and by tapping mode atomic force microscopy,⁵⁹ where combining VWF and APAC led to the structural alterations of VWF. In the case of intact endothelium, APAC only modestly interacted with PECAM (platelet endothelial cell adhesion molecule)-1 (CD31 [cluster of differentiation 31]) or podocalyxin.⁷⁰

APAC, when provided locally during vascular surgery or balloon injury, selectively targets subendothelial structures of VWF and laminins on arterial injury sites, unlike on intact endothelium.

Acute Arterial Thrombosis Models With Local or Systemic Administration of APAC

The antithrombotic efficacy of APAC was studied by the modified Folts model in baboons. Local administration of APAC strikingly inhibited occlusive thrombus formation at the site of crush injury and 30% to 90% stenosis during follow-up.⁵⁷ In contrast, UFH failed to prevent occlusion.⁵⁷

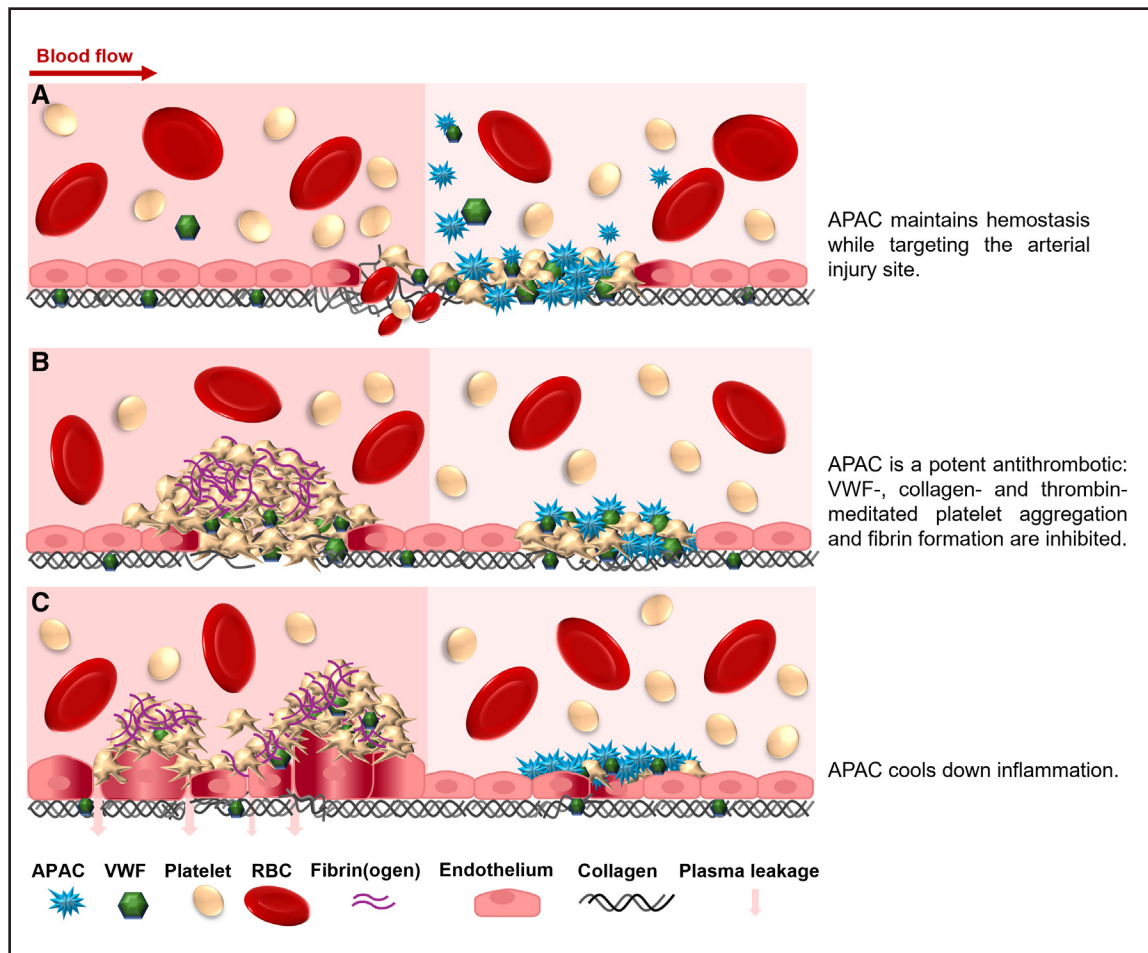


Figure 3. Mode of action of antiplatelet and anticoagulant (APAC).

Under arterial blood flow conditions, APAC (A) targets vascular injury sites and allows platelet adhesion, while preserving primary hemostasis,⁶⁸ (B) inhibits VWF (von Willebrand Factor)-mediated, collagen-mediated, and thrombin-mediated platelet aggregation and fibrin formation, being a potent antithrombotic,⁵⁷ and (C) cools down inflammation.^{60,61,69}

In the second thrombosis model in baboons without any systemic anticoagulation, a collagen-coated graft was incubated with APAC or UFH and connected to an extracorporeal arteriovenous shunt.⁵⁷ APAC reduced platelet deposition by 34% and fibrin accumulation by 45% to the shunt versus UFH. APAC also attenuated distal thrombus formation by 63%. Once again, the effects of UFH were similar to those of untreated controls.⁵⁷

Systemic administration of APAC also strongly inhibited arterial thrombosis and platelet deposition. In mice, APAC administered before the induction of carotid artery injury prolonged the time to thrombotic occlusion by 2-fold compared with UFH and vehicle.⁶⁸ APAC also depicted 50% less TF activity both in plasma and carotid lysates than UFH.⁶⁸

Moreover, in a surgical collagen-induced thrombosis model, where the collagen-rich adventitial surface of an inverted epigastric artery from a donor mouse was inserted into the recipient's carotid artery, providing both a thrombogenic surface and a narrowed lumen, APAC

prophylactically reduced platelet accumulation versus saline-treated mice.⁶⁸

Atherosclerosis With an ApoE^{-/-} Mouse Model

In an ApoE^{-/-} (apolipoprotein E knockout) mouse model of atherosclerosis, enhanced by collar-triggered stenosis on the carotid artery, APAC prevented early development of atherosclerosis, reducing the plaque volume by 60% from that of controls.⁶⁹ At maximal stenosis sites, APAC decreased the plaque size by over 50%. While the medial layer remained unaffected, APAC decreased macrophage content, while their proportion relative to lesion size remained unchanged.⁶⁹ Collagen and necrotic areas also diminished by over 50% in the APAC-treated mice, unlike the vascular smooth muscle cells.⁶⁹

When APAC was administered 2.5 weeks after the initiation of atherosclerosis, the carotid plaque at maximal stenosis or average lesion size along the plaque did not change, but plaque volume tended to decrease.⁶⁹ The collagen content was unaffected, but APAC diminished

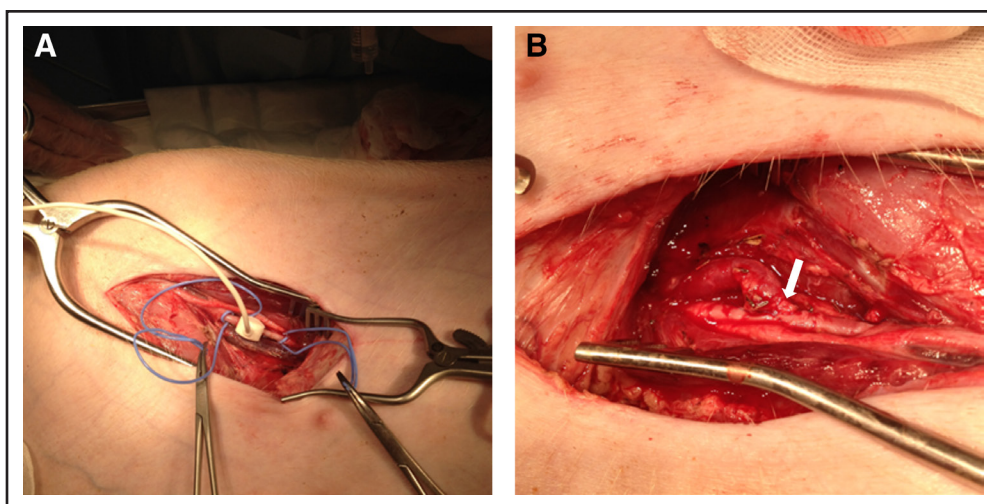


Figure 4. Administration of antiplatelet and anticoagulant (APAC) after creation of arteriovenous fistula (AVF) between the femoral artery and vein in the pig hind leg.

A, Smooth clamping of the side-to-end anastomosis of femoral AVF to administer and incubate APAC to the vessel surface for 2 minutes before releasing blood flow. **B**, AVF (white arrow) under blood flow before excision and assessment of colocalization of VWF (von Willebrand Factor) and APAC using immunohistochemistry. Photographs are unpublished but relate to Barreiro et al.⁷⁰

the necrotic area by 25%,⁶⁹ referring to improved lesion stability during progression.

In the more advanced atherosclerotic plaques, APAC reduced inflammation, as macrophage area was decreased by 42%, coinciding with reduced expression of tumor necrosis factor α mRNA.⁶⁹ However, APAC did not affect macrophage phenotype or other inflammatory

cytokines, that is, CCL2 (C-C motif chemokine ligand 2), IL (interleukin)-1 β , and IL-6.⁶⁹

In all, in the ApoE^{-/-} mouse model with hemodynamically relevant atherosclerosis, APAC clearly alleviated lesion formation and its thrombogenic properties. In the progressing plaques, APAC appeared to lessen inflammation and support lesion stability.

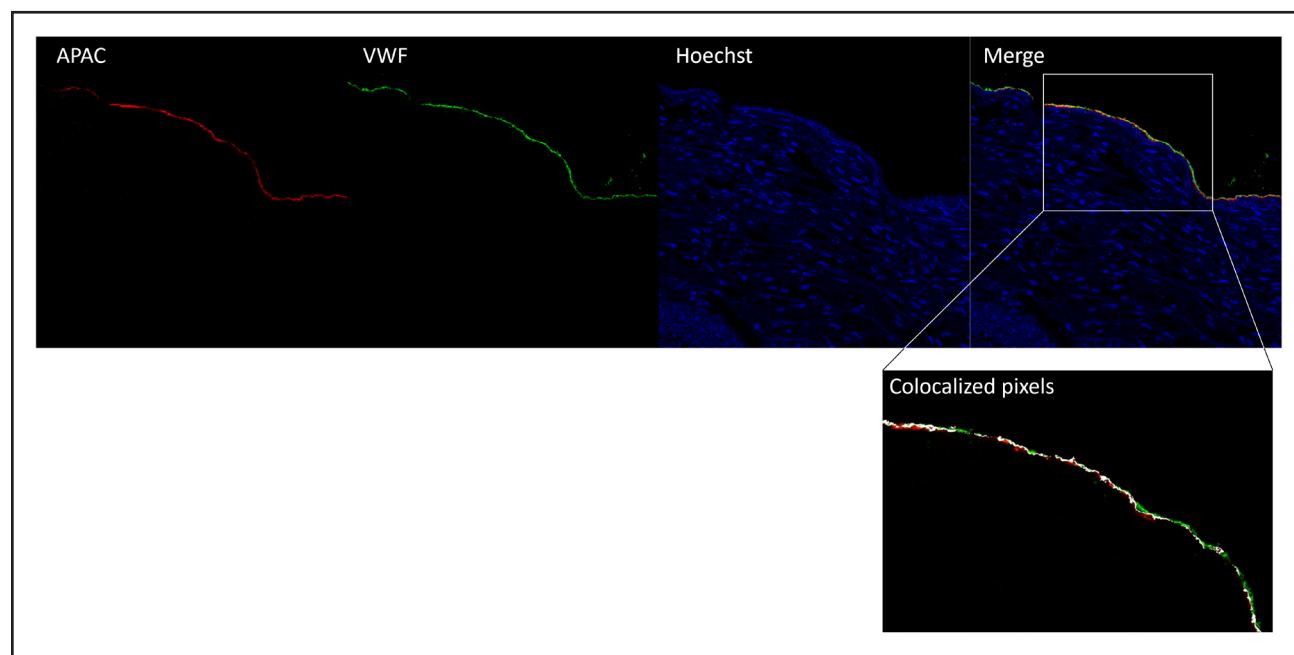


Figure 5. Antiplatelet and anticoagulant (APAC) targets the balloon-denuded arterial site and colocalizes with VWF (von Willebrand Factor) in a pig model.

After balloon injury of the artery, APAC-biotin (1.5 mg) was administered by a diffusion balloon catheter for 2 minutes, then the balloon was removed, and circulation was restored for 30 minutes before tissue harvesting. Tissues were immuno-stained for APAC (red), VWF (green), and nuclei by Hoechst (blue). APAC and VWF were detected on the luminal side of the artery. The merged and colocalized pixels of APAC and VWF are shown. Pictures are unpublished but relate to Barreiro et al.⁷⁰

IRI Model in Rat Kidneys

In a preventive kidney IRI model in rats, APAC was renoprotective both in reversible and irreversible acute kidney injuries. APAC decreased serum NGAL (neutrophil gelatinase-associated lipocalin) levels, an acute kidney injury biomarker.⁶¹ Histopathologic analysis revealed reduced hyaluronan, a regulator of innate immunity, and Kim-1 (kidney injury molecule-1) density, a specific marker for proximal tubular damage.⁶¹ At 3 days post-IRI, tubular dilation, epithelial necrosis, flattening, and casts confirmed severe tubular injury caused by reversible IRI (30 minutes) in the controls, while APAC-treated animals were rescued.⁶¹ Unlike APAC, UFH did not decrease innate or tubulointerstitial injury; UFH-treatment resembled the control group.⁶¹

In an irreversible kidney IRI model (60 minutes), all APAC-treated rats had renal recovery and survived, unlike 25% of the controls.⁶¹ Moreover, APAC reduced molecular markers of vascular destabilization, proinflammation (angiopoietin-2, syndecan-1), and kidney injury (creatinine, urea nitrogen, and NGAL).⁶¹ In all, APAC alleviated the innate kidney injury.

Ischemic Stroke Model in Mice

Ischemic stroke was induced in mice by a transient occlusion of the right middle cerebral artery. Prophylactic administration of APAC or UFH reduced cerebral IRI and improved neurological outcomes, without introducing cerebral hemorrhaging.⁶² However, this effect was lost when mice were treated after the onset of ischemia.⁶² The clinical potential of targeted platelet and coagulation inhibition would be relevant during neurosurgery, where it is critical to avoid both local thrombosis and bleeding complications through the systemic long-term platelet inhibition of the current regimen.

Pharmacodynamic Assessments by Functional Analysis

A direct method to detect systemic exposure of APAC concentrations *ex vivo* is currently unavailable. Instead, functional assays, routinely clinically used to monitor heparin and platelet function in humans, can be used as surrogate pharmacodynamic markers (eg, aPTT and thrombin time in plasma, activated clotting time, and InTEM in blood).

In rats, aPTT remained at baseline at 10 minutes after small intravenous doses of APAC (0.06 and 0.13 mg/kg) but prolonged 2-fold at 0.32 mg/kg.⁷⁰ Based on ⁶⁴Cu-labeled APAC, T_{1/2} was only ≈15 minutes up to 0.5 mg/kg but prolonged to ≈60 minutes at the high dose of 7.3 mg/kg.⁶¹ APAC was eliminated by the liver and kidneys, as heparin.^{57,61} Positron emission tomography imaging revealed early accumulation predominantly in the liver, spleen, and kidneys.⁶¹

In pigs, escalating intravenous doses prolonged thrombin time, aPTT, activated clotting time, and

rotational thromboelastometry InTEM (CT and CFT) by both APAC and UFH (0.15–1.5 mg/kg).^{59,71} The anticoagulant effects of APAC were transient and reverted to baseline within 75 minutes.⁷¹

TOXICOLOGY AND SAFETY EVALUATION

Safety

Intravenous administration of APAC did not impact blood cell counts nor induced spontaneous bleeding or hemorrhagic complications in rats, mice, and primates.^{57,61,67,69}

The reversal of APAC activity with protamine sulphate in human and porcine plasma and *in vivo* in pigs was confirmed.^{57,71} Interestingly, administration of protamine sulphate in pigs reversed the effect of APAC, whereas it was ineffective at the same dose of UFH.⁷¹

Toxicology

The no-observed-adverse-effect level of APAC was 20 mg/kg in rats and 3 mg/kg in primates by a single intravenous infusion. The no-observed-adverse-effect level for daily APAC for 14 days was established at 10 mg/kg per day for rats and 0.5 mg/kg per day for primates.⁶⁷ Importantly, this repeated dosing did not accumulate or impair tolerance, while aPTT profiles returned to baseline within 6 hours in both species.⁶⁷ In addition, the repeated dosing in pigs did not induce thrombocytopenia,⁷¹ which was confirmed in a phase I study with repeated APAC dosing in healthy participants (CRSC20007; unpublished).

EARLY CLINICAL EXPERIENCE OF APAC AND NEXT STEPS

Based on preclinical studies, 0.5 mg/kg intravenous was selected as the maximal dose when assessing the dose-dependent safety in healthy participants (CRSC20007). For AVF surgery, tissue safety, and healing, the local doses leading to 0.75- to 4.5-mg APAC exposure at the AVF surgery site (CRSC19001) have been completed. Both studies showed clinical safety (unpublished), the data being reported to authorities for phase 2 studies. The indications include maturation and patency of AVF (submitted) and PAOD (NCT07352800) in association with balloon angioplasty and, in cases where intervention cannot be done, re-exposure to APAC aiming at alleviating the disease progression. In addition, ⁸⁹Zr-labeled APAC for positron emission tomography aims at detecting arterial lesions in PAOD (NCT06204237).

In all, this APAC has undergone significant steps toward the development of a novel antithrombotic with dual antiplatelet and anticoagulant potential, with a local arterial injury-targeting profile and alleviation of inflammation, which hopefully will respond to clinical needs.

ARTICLE INFORMATION

Received January 15, 2026; accepted May 13, 2026.

Affiliations

Research Program in Systems Oncology, Faculty of Medicine, University of Helsinki, Finland (A.Y.K., A.J., R.L.). Division of Anesthesiology, Department of Anesthesiology, Intensive Care and Pain Medicine, Helsinki University Hospital, Finland (H.P.). Clinical Research Institute HUCH, Helsinki, Finland (A.Y.K., A.J.). Coagulation Disorders Unit, Department of Hematology, Comprehensive Cancer Center, Helsinki University Hospital, Finland (H.P., R.L.). Aplagon Ltd, Helsinki, Finland (R.L.).

Acknowledgments

The authors gratefully acknowledge the excellent collaboration provided by many international coworkers cited in this review. They acknowledge laboratory technician Marja Lemponen for many years of cooperation and Karina Barreiro, PhD, for Figure 5.

Sources of Funding

This work was supported by Aplagon Ltd and the European Innovation Council (EIC grant 190168043). The manufacturing of antiplatelet and anticoagulant was performed by Cadila Pharmaceuticals, India.

Disclosures

R. Lassila is the Chief Medical Officer and the Chief Scientific Officer of Aplagon Ltd, Helsinki, Finland. A.Y. Kern and A. Jouppila received research funding via Clinical Research Institute HUCH from Aplagon Ltd. H. Pitkänen reports no conflicts.

REFERENCES

- Sakariassen KS, Orning L, Turitto VT. The impact of blood shear rate on arterial thrombus formation. *Future Sci OA*. 2015;1:FSO30. doi: 10.4155/fso.15.28
- Panteleev MA, Korin N, Reesink KD, Bark DL, Cosemans JM, Gardiner EE, Mangin PH. Wall shear rates in human and mouse arteries: standardization of hemodynamics in vitro blood flow assays: communication from the ISTH SSC subcommittee on biorheology. *J Thromb Haemost*. 2021;19:588–595. doi: 10.1111/jth.15174
- Stroev PV, Hoskins PR, Easson WJ. Distribution of wall shear rate throughout the arterial tree: a case study. *Atherosclerosis*. 2007;191:276–280. doi: 10.1016/j.atherosclerosis.2006.05.029
- Turitto VT, Weiss HJ. Red blood cells: their dual role in thrombus formation. *Science*. 1980;207:541–543. doi: 10.1126/science.7352265
- Houme C, Huang Z, Shankar KN, Young SJ, Roberts ME, Diamond SL, Tomaiuolo M, Stalker TJ, Lu L, Sinno T. From imaging to computational domains for physics-driven molecular biology simulations: hindered diffusion in platelet masses. *PLoS Comput Biol*. 2025;21(7):2025.02.06.636817. doi: 10.1371/journal.pcbi.1012853
- Savage B, Saldivar E, Ruggeri ZM. Initiation of platelet adhesion by arrest onto fibrinogen or translocation on von Willebrand Factor. *Cell*. 1996;84:289–297. doi: 10.1016/s0092-8674(00)80983-6
- Liu Y, Crossen J, Stalker TJ, Diamond SL. Fluorescent peptide for detecting factor XIIIa activity and fibrin in whole blood clots forming under flow. *Res Pract Thromb Haemost*. 2024;8:102291. doi: 10.1016/j.rpth.2023.102291
- Cines DB, Lebedeva T, Nagaswami C, Hayes V, Masefski W, Litvinov RI, Rauova L, Lowery TJ, Weisel JW. Clot contraction: compression of erythrocytes into tightly packed polyhedra and redistribution of platelets and fibrin. *Blood*. 2014;123:1596–1603. doi: 10.1182/blood-2013-08-523860
- Lassila R, Weisel JW. Role of red blood cells in clinically relevant bleeding disorders and complications. *J Thromb Haemost*. 2023;21:3024–3032. doi: 10.1016/j.jth.2023.05.009
- Litvinov RI, Weisel JW. Role of red blood cells in haemostasis and thrombosis. *ISBT Sci Ser*. 2017;12:176–183. doi: 10.1111/voxs.12331
- Chernysh IN, Nagaswami C, Kosolapova S, Peshkova AD, Cuker A, Cines DB, Cambor CL, Litvinov RI, Weisel JW. The distinctive structure and composition of arterial and venous thrombi and pulmonary emboli. *Sci Rep*. 2020;10:5112. doi: 10.1038/s41598-020-59526-x
- Siljander P, Carpen O, Lassila R. Platelet-derived microparticles associate with fibrin during thrombosis. *Blood*. 1996;87:4651–4663. doi: 10.1182/blood.V87.11.4651.bloodjournal87114651
- Clancy JW, Schmidtman M, D'Souza-Schorey C. The ins and outs of microvesicles. *FASEB Bioadv*. 2021;3:399–406. doi: 10.1096/fba.2020-00127
- Grover SP, Mackman N. Tissue factor in atherosclerosis and atherothrombosis. *Atherosclerosis*. 2020;307:80–86. doi: 10.1016/j.atherosclerosis.2020.06.003
- Lentz BR. Exposure of platelet membrane phosphatidylserine regulates blood coagulation. *Prog Lipid Res*. 2003;42:423–438. doi: 10.1016/s0163-7827(03)00025-0
- Reddy EC, Rand ML. Procoagulant phosphatidylserine-exposing platelets in vitro and in vivo. *Front Cardiovasc Med*. 2020;7:15. doi: 10.3389/fcvm.2020.00015
- Neubauer K, Zieger B. Endothelial cells and coagulation. *Cell Tissue Res*. 2022;387:391–398. doi: 10.1007/s00441-021-03471-2
- Kohli S, Shahzad K, Jouppila A, Holthöfer H, Isermann B, Lassila R. Thrombosis and inflammation—a dynamic interplay and the role of glycosaminoglycans and activated protein C. *Front Cardiovasc Med*. 2022;9:866751. doi: 10.3389/fcvm.2022.866751
- Ferreira G, Taylor A, Mensah SA. Deciphering the triad of endothelial glycocalyx, von Willebrand Factor, and P-selectin in inflammation-induced coagulation. *Front Cell Dev Biol*. 2024;12:1372355. doi: 10.3389/fcell.2024.1372355
- Jackson SP, Darbousset R, Schoenwaelder SM. Thromboinflammation: challenges of therapeutically targeting coagulation and other host defense mechanisms. *Blood*. 2019;133:906–918. doi: 10.1182/blood-2018-11-882993
- Vlodavsky I, Ilan N, Nadir Y, Brenner B, Katz BZ, Naggi A, Torri G, Casu B, Sasisekharan R. Heparanase, heparin and the coagulation system in cancer progression. *Thromb Res*. 2007;120:S112–S120. doi: 10.1016/S0049-3848(07)70139-1
- Angiolillo DJ, Rollini F, Storey RF, Bhatt DL, James S, Schneider DJ, Sibbing D, So DYF, Trenk D, Alexopoulos D, et al. International expert consensus on switching platelet P2Y12 receptor-inhibiting therapies. *Circulation*. 2017;136:1955–1975. doi: 10.1161/CIRCULATIONAHA.117031164
- Eikelboom JW, Hirsh J, Weitz JI, Johnston M, Yi Q, Yusuf S. Aspirin-resistant thromboxane biosynthesis and the risk of myocardial infarction, stroke, or cardiovascular death in patients at high risk for cardiovascular events. *Circulation*. 2002;105:1650–1655. doi: 10.1161/01.cir.0000013777.21160.07
- Gum PA, Kottke-Marchant K, Welsh PA, White J, Topol EJ. A prospective, blinded determination of the natural history of aspirin resistance among stable patients with cardiovascular disease. *J Am Coll Cardiol*. 2003;41:961–965. doi: 10.1016/s0735-1097(02)03014-0
- Matetzky S, Shenkman B, Guetta V, Shechter M, Beinart R, Goldenberg I, Novikov I, Pres H, Savion N, Varon D, et al. Clopidogrel resistance is associated with increased risk of recurrent atherothrombotic events in patients with acute myocardial infarction. *Circulation*. 2004;109:3171–3175. doi: 10.1161/01.CIR.0000130846.46168.03
- Ray S. Clopidogrel resistance: the way forward. *Indian Heart J*. 2014;66:530–534. doi: 10.1016/j.ihj.2014.08.012
- Gasparyan AY. Aspirin and clopidogrel resistance: methodological challenges and opportunities. *Vasc Health Risk Manag*. 2010;6:109–112. doi: 10.2147/vhrm.s9087
- Lepäntalo A, Virtanen KS, Heikkilä J, Wartiovaara U, Lassila R. Limited early antiplatelet effect of 300 mg clopidogrel in patients with aspirin therapy undergoing percutaneous coronary interventions. *Eur Heart J*. 2004;25:476–483. doi: 10.1016/j.ehj.2003.12.016
- Lepäntalo A, Mikkelsson J, Reséndiz JC, Viiri L, Backman JT, Kankuri E, Karhunen PJ, Lassila R. Polymorphisms of COX-1 and GP VI associate with the antiplatelet effect of aspirin in coronary artery disease patients. *Thromb Haemost*. 2006;95:253–259. doi: 10.1160/TH05-07-0516
- Urban P, Mehran R, Collieran R, Angiolillo DJ, Byrne RA, Capodanno D, Cuisset T, Cutlip D, Eerdmans P, Eikelboom J, et al. Defining high bleeding risk in patients undergoing percutaneous coronary intervention. *Circulation*. 2019;140:240–261. doi: 10.1161/CIRCULATIONAHA.119.040167
- Schutgens REG, Jimenez-Yuste V, Escobar M, Falanga A, Gigante B, Klamroth R, Lassila R, Leebeek FWG, Makris M, Owaidah T, et al. Anti-thrombotic treatment in patients with hemophilia: an EHA-ISTH-EAHAD-ESO clinical practice guidance. *Hemasphere*. 2023;7:e900. doi: 10.1097/HS9.0000000000000900
- Jeppsson A, James S, Moller CH, Malm CJ, Dalén M, Vanky F, Modrau IS, Andersen K, Anttila V, Atroshchenko GV, et al; TACS Trial Group. Ticagrelor and aspirin or aspirin alone after coronary surgery for acute coronary syndrome. *N Engl J Med*. 2025;393:2313–2323. doi: 10.1056/NEJMoa2508026
- Bouma BN, Meijers JCM. Thrombin-activatable fibrinolysis inhibitor (TAFI), plasma procarboxypeptidase B, procarboxypeptidase R, procarboxypeptidase U. *J Thromb Haemost*. 2003;1:1566–1574. doi: 10.1046/j.1538-7836.2003.00329.x

34. Lassila R, Peltonen S, Lepäntalo M, Saarinen O, Kauhanen P, Manninen V. Severity of peripheral atherosclerosis is associated with fibrinogen and degradation of cross-linked fibrin. *Arterioscler Thromb*. 1993;13:1738–1742. doi: 10.1161/01.atv.13.12.1738
35. Eikelboom JW, Connolly SJ, Bosch J, Dagenais GR, Hart RG, Shestakovska O, Diaz R, Alings M, Lonn EM, Anand SS, et al; COMPASS Investigators. Rivaroxaban with or without aspirin in stable cardiovascular disease. *N Engl J Med*. 2017;377:1319–1330. doi: 10.1056/NEJMoa1709118
36. Bonaca MP, Bauersachs RM, Anand SS, Debus ES, Nehler MR, Patel MR, Fanelli F, Capell WH, Diao L, Jaeger N, et al. Rivaroxaban in peripheral artery disease after revascularization. *N Engl J Med*. 2020;382:1994–2004. doi: 10.1056/NEJMoa2000052
37. De Moerloose P, Boehlen F. Inherited thrombophilia in arterial disease: a selective review. *Semin Hematol*. 2007;44:106–113. doi: 10.1053/j.seminhematol.2007.01.008
38. VanderPlum CJ, Cantor RS, Machado D, Boyle G, May L, Griffiths E, Niebler RA, Lorts A, Rossano J, Sutcliffe DL, et al. Utilization and outcomes of children treated with direct thrombin inhibitors on paracorporeal ventricular assist device support. *ASAIO J*. 2020;66:939–945. doi: 10.1097/MAT.0000000000001093
39. Bembea MM, Annich G, Rycus P, Oldenburg G, Berkowitz I, Pronovost P. Variability in anticoagulation management of patients on extracorporeal membrane oxygenation: an international survey. *Pediatr Crit Care Med*. 2013;14:e77–e84. doi: 10.1097/PCC.0b013e31827127e4
40. Garcia DA, Baglin TP, Weitz JI, Samama MM. Parenteral Anticoagulants: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012;141:e24S–e43S. doi: 10.1378/chest.11-2291
41. Newall F, Johnston L, Ignjatovic V, Monagle P. Unfractionated heparin therapy in infants and children. *Pediatrics*. 2009;123:e510–e518. doi: 10.1542/peds.2008-2052
42. Oliver WC. Anticoagulation and coagulation management for ECMO. *Semin Cardiothorac Vasc Anesth*. 2009;13:154–175. doi: 10.1177/1089253209347384
43. Badimon L, Badimon J, Lassila R, Heras M, Chesebro J, Fuster V. Thrombin regulation of platelet interaction with damaged vessel wall and isolated collagen type I at arterial flow conditions in a porcine model: effects of hirudins, heparin, and calcium chelation. *Blood*. 1991;78:423–434. doi: 10.1182/blood.v78.2.423.423
44. Van Den Helm S, Sparks CN, Ignjatovic V, Monagle P, Attard C. Increased risk for thromboembolism after Fontan surgery: considerations for thromboprophylaxis. *Front Pediatr*. 2022;10:803408. doi: 10.3389/fped.2022.803408
45. Gewillig M, Brown SC, van de Bruaene A, Rychik J. Providing a framework of principles for conceptualising the Fontan circulation. *Acta Paediatr*. 2020;109:651–658. doi: 10.1111/apa.15098
46. Sasikumar N, Hermuzi A, Fan CPS, Lee KJ, Chaturvedi R, Hickey E, Honjo O, Van Arsdell GS, Caldaroni CA, Agarwal A, et al. Outcomes of Blalock-Taussig shunts in current era: a single center experience. *Congenit Heart Dis*. 2017;12:808–814. doi: 10.1111/chd.12516
47. Şişli E, Tuncer ON, Şenkaya S, Doğan E, Şahin H, Ayık MF, Atay Y. Blalock-Taussig shunt size: should it be based on body weight or target branch pulmonary artery size? *Pediatr Cardiol*. 2019;40:38–44. doi: 10.1007/s00246-018-1958-9
48. Headrick AT, Qureshi AM, Ghanayem NS, Heinle J, Anders M. In-hospital morbidity and mortality after modified Blalock-Taussig-Thomas shunts. *Ann Thorac Surg*. 2022;114:168–175. doi: 10.1016/j.athoracsurg.2021.11.003
49. Alsagheir A, Koziaz A, Makhdoum A, Contreras J, Alraddadi H, Abdalla T, Benson L, Chaturvedi RR, Honjo O. Duct stenting versus modified Blalock-Taussig shunt in neonates and infants with duct-dependent pulmonary blood flow: a systematic review and meta-analysis. *J Thorac Cardiovasc Surg*. 2021;161:379–390.e8. doi: 10.1016/j.jtcvs.2020.06.008
50. Puurunen M, Lassila R. Successful treatment of mesenteric venous thrombosis with recombinant hirudin—a report of five cases. *Thromb Res*. 2006;118:241–245. doi: 10.1016/j.thromres.2005.06.003
51. Salmela B, Albäck A, Räike P, Lepäntalo M, Lassila R. A direct thrombin inhibitor, lepirudin, for thrombophilic patients with inoperable critical limb ischemia. *Thromb Res*. 2009;123:719–723. doi: 10.1016/j.thromres.2008.11.015
52. Salmela B, Nordin A, Vuoristo M, Mäksälä H, Numminen K, Lassila R. Budd-Chiari syndrome in a young female with factor V Leiden mutation: successful treatment with lepirudin, a direct thrombin inhibitor. *Thromb Res*. 2008;121:769–772. doi: 10.1016/j.thromres.2007.08.008
53. Haapaniemi E, Helenius J, Jakovljević D, Soinne L, Syrjäälä M, Kaste M, Lassila R, Tatlisumak T. Ischaemic stroke patients with heterozygous factor V Leiden present with multiple brain infarctions and widespread atherothrombotic disease. *Thromb Haemost*. 2009;101:145–150. doi: 10.1160/TH07-12-0754
54. Kauhanen P, Kovanen PT, Lassila R. Coimmobilized native macromolecular heparin proteoglycans strongly inhibit platelet-collagen interactions in flowing blood. *Arterioscler Thromb Vasc Biol*. 2000;20:E113–E119. doi: 10.1161/01.atv.20.11.e113
55. Lassila R, Lindstedt K, Kovanen PT. Native macromolecular heparin proteoglycans exocytosed from stimulated rat serosal mast cells strongly inhibit platelet-collagen interactions. *Arterioscler Thromb Vasc Biol*. 1997;17:3578–3587. doi: 10.1161/01.atv.17.12.3578
56. Olsson E, Asko-Seljavaara S, Lassila R. Topically administered macromolecular heparin proteoglycans inhibit thrombus growth in microvascular anastomoses. *Thromb Haemost*. 2002;87:245–251.
57. Lassila R, Jouppila A. Mast cell-derived heparin proteoglycans as a model for a local antithrombotic. *Semin Thromb Hemost*. 2014;40:837–844. doi: 10.1055/s-0034-1395157
58. Jouppila A, Nevola I, Lemponen M, Mattila T, Lassila R. Antiplatelet-anticoagulant, APAC, a mimic of endogenous heparin, is an antithrombotic with von Willebrand Factor-mediated characteristics. *Thromb Res*. 2025;250:109318. doi: 10.1016/j.thromres.2025.109318
59. Winzely M, Jouppila A, Ramer G, Lux L, Lendl B, Barreiro K, Lassila R, Friedbacher G. AFM investigation of APAC (antiplatelet and anticoagulant heparin proteoglycan). *Anal Bioanal Chem*. 2022;414:1029–1038. doi: 10.1007/s00216-021-03765-y
60. Kotimaa J, Rezola MA, Jouppila A, Lassila R, Meri S. Interactions of antiplatelet, anticoagulant (APAC) with C1q and FH modulate complement activation on injured cells. *Blood Vessels Thromb Hemost*. 2026;3:100150. doi: 10.1016/j.bvth.2026.100150
61. Tuuminen R, Jouppila A, Salvail D, Laurent CE, Benoit MC, Syrjäälä S, Helin H, Lemström K, Lassila R. Dual antiplatelet and anticoagulant APAC prevents experimental ischemia-reperfusion-induced acute kidney injury. *Clin Exp Nephrol*. 2017;21:436–445. doi: 10.1007/s10157-016-1308-2
62. Denorme F, Frösen J, Jouppila A, Lindgren A, Resendiz-Nieves JC, Manninen H, De Meyer SF, Lassila R. Pretreatment with a dual antiplatelet and anticoagulant (APAC) reduces ischemia-reperfusion injury in a mouse model of temporary middle cerebral artery occlusion-implications for neurovascular procedures. *Acta Neurochir (Wien)*. 2024;166:137. doi: 10.1007/s00701-024-06017-x
63. Chen J, Verni CC, Jouppila A, Lassila R, Diamond SL. Dual antiplatelet and anticoagulant (APAC) heparin proteoglycan mimetic with shear-dependent effects on platelet-collagen binding and thrombin generation. *Thromb Res*. 2018;169:143–151. doi: 10.1016/j.thromres.2018.07.026
64. Vögtle T, Sharma S, Mori J, Nagy Z, Semenjak D, Scandola C, Geer MJ, Smith CW, Lane J, Pollack S, et al. Heparan sulfates are critical regulators of the inhibitory megakaryocyte-platelet receptor G6b-B. *Elife*. 2019;8:e46840. doi: 10.7554/eLife.46840
65. Shi D, Zhao H, Bu C, Fraser K, Wang H, Dordick JS, Linhardt RJ, Zhang F, Shi F, Chi L. New insights into the binding of PF4 to long heparin oligosaccharides in ultralarge complexes using mass spectrometry. *J Thromb Haemost*. 2023;21:3608–3618. doi: 10.1016/j.jth.2023.08.020
66. Villalobos D, Wen R, Wang D, Witman N. Novel PF4/heparin complexes drive platelet activation in heparin-induced thrombocytopenia. *Blood*. 2025;146:3097–3097. doi: 10.1182/blood-2025-3097
67. Craige S, Jouppila A, Humphries B, Lassila R. Safety and functional pharmacokinetic profile of APAC, a novel intravascular antiplatelet and anticoagulant. *J Cardiovasc Pharmacol*. 2021;78:453–462. doi: 10.1097/FJC.0000000000001080
68. Bonetti NR, Jouppila AS, Saeedi Saravi SS, Cooley BC, Pasterk L, Liberale LL, Gobbato S, Lüscher TF, Camici GG, Lassila RP, et al. Intravenously administered APAC, a dual AntiPlatelet AntiCoagulant, targets arterial injury site to inhibit platelet thrombus formation and tissue factor activity in mice. *Thromb Res*. 2023;228:163–171. doi: 10.1016/j.thromres.2023.04.010
69. Bot I, Delfos L, Hemme E, Bernabé Kleijn MNA, Van Santbrink PJ, Foks AC, Kovanen PT, Jouppila A, Lassila R. Treatment with APAC, a dual antiplatelet anticoagulant heparin proteoglycan mimetic, limits early collar-induced carotid atherosclerotic plaque development in ApoE^{-/-} mice. *Atherosclerosis*. 2024;397:118567. doi: 10.1016/j.atherosclerosis.2024.118567
70. Barreiro KA, Tulamo R, Jouppila A, Albäck A, Lassila R. Novel locally acting dual antiplatelet and anticoagulant (APAC) targets multiple sites of vascular injury in an experimental porcine model. *Eur J Vasc Endovasc Surg*. 2019;58:903–911. doi: 10.1016/j.ejvs.2019.05.019
71. Bagoly Z, Jouppila A, Orbán-Kálmándi R, Lóczi L, Bomberák D, Kádár ZA, Deák A, Mátrai A, Debreceni IB, Kappelmayer J, et al. Escalating doses of intravenous APAC demonstrate antithrombotic effect in pigs. *Thromb J*. 2025;23:57. doi: 10.1186/s12959-025-00742-8